## Communication, Rating, and Dispute Reporting

1. **Communication**

***Preset Messaging***

Preset messages will cover common positive and negative use cases for both Seekers and Givers:

**Positive Use Cases:**

- **Seeker:**
- "I have arrived at the parking spot."

- **Giver:**
- "The parking spot is ready for you."
- "The parking spot is secure and clean."

**Negative Use Cases:**

- **Seeker:**
- "I am delayed but will arrive very soon."
- "Unexpected delay occurred; Can you extend for 5 minutes?"
- "I have arrived earlier than expected."
- "I am waiting near the parking spot for now."
- **Giver:**
- "I am delayed and need 5 more minutes."
- "I will wait until the hosting period ends; if you don't arrive, I will leave the spot."
- "I am waiting in my vehicle until the swap is completed."
- "I am in my car, but I need a few more minutes before I can leave due to personal reasons."
- **Customization:** Preset messages are non-customizable to maintain clarity and usability across users.

- **Communication Method:** Preset messages are available in-app and can be accessed from the "Communications" tab during active matches. These messages are integrated directly into the app's user interface, ensuring they are readily accessible at critical moments, such as during parking handovers or disputes. The implementation involves:
- **Tab Design:** A dedicated "Communications" tab within active matches, prominently displayed for ease of access.
- **Preset Message Categories:** Organized into positive and negative use cases for better context-based selection.
- **Quick Access Options:** Users can select messages with a single tap, minimizing response time and ensuring clarity during interactions.
- **Call Feature**
- **After Match:** Users are presented with a call button.
- **Functionality:** The call button will dial the mobile number associated with the user's account registration.
- If the user registered via email, the call button will not appear.
- **Customization:** Users can toggle this feature on or off in their profile settings.

Example Usage:

- A Seeker experiencing delays can quickly select and send: "I am delayed but will arrive in 5 minutes." This ensures timely communication without the need for typing.

2. **Rating System**

***System Design***

- **Star Rating:** Users can rate the other party on a scale of 1 to 5 stars.
- **Visibility:** Ratings are visible to the rated party, including the identity of the user providing the rating.

***Rating Triggers***

- **After Successful Match Completion:** Users are prompted to rate each other immediately after the parking swap. A notification appears prompting both parties to leave a star rating and optional feedback.

- **Missed Matches:** If a swap fails and a dispute occurs, ratings are delayed until dispute resolution. Specifically:
- Once the resolution outcome has been finalized, either through automated or manual review, users are notified that they can now provide their ratings.

***Historical Logs***

- Ratings and feedback are stored for as long as the user account is active. This approach ensures a comprehensive history of user interactions, fostering transparency and trust within the community:
- **Long-Term Reputation Management:** Retaining ratings and feedback indefinitely allows users to build a consistent and reliable profile, helping others make informed decisions about engaging with them.
- **Support for Dispute Resolutions:** Historical data provides critical context for resolving disputes, even if they arise months or years later.

3. **Dispute Reporting**

***Match Details***

The "Match Details" page serves as a centralized interface where users can access critical information related to a specific parking interaction. It includes:

- **Context-Specific:** Focuses on a single, active parking interaction.
- **Content:** Displays real-time transaction details, communication logs, and status updates relevant to a specific match.
- **Purpose:** Provides immediate and actionable information during or shortly after the interaction.
- **Availability:** Accessible only during active or recently completed matches.

***Key Features:***

- Real-time updates.
- Dispute submission for the specific interaction.
- Prompts for feedback and ratings upon completion.

***Dispute Submission***

Users can initiate a dispute directly from this page with predefined reasons and optional comments for additional context.

- Users can initiate a dispute The "Match Details" page serves as a centralized interface where users can access critical information related to a specific parking interaction. It includes:
- **Transaction Summary:** Displays details of the parking arrangement, such as the agreed time, location, and token transactions.
- **Communication Logs:** Shows preset messages exchanged between the Seeker and Giver during the active match.
- **Dispute Submission:** Provides an option for users to submit a dispute with predefined reasons and additional comments.
- **Live Updates:** Offers real-time updates on the parking status, such as arrival confirmations or delays.
- **Rating and Feedback:** Prompts users to leave ratings and optional feedback after the match is completed or resolved.

The page design ensures all relevant information is easily accessible, improving transparency and user experience. 12 hours of the incident.

- Common dispute reasons:

**Resolved via Automated Resolution**

- **Seeker did not show up:** Geolocation and timestamp logs can confirm if the Seeker failed to arrive at the designated spot within the agreed time.
- **Giver was not present:** Geolocation and timestamp logs can verify if the Giver was absent during the hosting period.
- **Seeker arrived late, exceeding the hosting duration:** Timestamps can determine if the Seeker's arrival was outside the agreed timeframe.
- **Giver left the spot before the agreed time:** Geolocation and timestamps can confirm if the Giver vacated the spot prematurely.

**Requires Manual Review**

- **Spot was occupied or inaccessible:** Verification requires additional evidence, such as photos or third-party confirmation.
- **The parking spot details were inaccurate:** Manual review is needed to assess discrepancies in the listing details.

- **Parking spot access was blocked or restricted:** Further investigation, possibly involving third-party validation, is necessary.
- **Host failed to confirm the swap completion:** Requires context from user activity logs and additional input from both parties.
- **User behavior was inappropriate or disrespectful:** Manual assessment is essential as this involves subjective judgment and potential evidence submission.

***Automated Resolution***

- **Resolution Criteria:**
- Geolocation data to verify arrival and presence. This involves using GPS logs to confirm the physical location of the Seeker and Giver during the hosting period.
- Timestamp logs to verify adherence to timing. The system checks whether both parties adhered to the agreed timeframe for parking handover.
- Consistency checks: The system compares activity logs, such as user interactions (e.g., "arrival" or "ready" confirmations), against reported disputes.
- **Resolution Process:**
- The system cross-references geolocation and timestamp data to confirm if the Seeker reached the designated spot and whether the Giver's hosting duration matched the agreed timeframe.
- If the data matches the expected behavior for both parties, the system automatically resolves the dispute in favor of the compliant user.
- In cases where data is ambiguous or incomplete, the system flags the dispute for manual review.
- **Resolution Timeline:**
- Automated resolution attempts are completed within 24 hours. This ensures timely feedback for straightforward disputes while allowing for rapid escalation if needed.

***Manual Review***

- **Escalation:** Disputes unresolved by the system within 24 hours are escalated to human review.
- **Resolution Timeline:** Manual reviews are completed within an additional 48 hours.

***Resolution Outcomes:***

- **Tokens Refunded:** In valid disputes, tokens are refunded to the appropriate party.

- **No MVP Penalties:** Repeated disputes will not impact user prioritization for MVP.

4. **Notifications**

***Notification Triggers***

**Onboarding:**

- Welcome message and instructions to complete phone verification. Example: "Welcome to [App Name]! Please verify your mobile number or email to start using the app."
- Reminder to complete full verification (vehicle details). Example: "Complete your vehicle verification to access full features like parking hosting and seeking."

**Parking Hosting:**

- Confirmation of hosting start. Example: "Your parking spot at [Location] is now active and available for seekers."
- Notification of seeker match. Example: "A seeker has matched with your parking spot. They will arrive shortly."
- Reminder when hosting period is about to end. Example: "Your hosting period is about to expire. Extend the time if needed."

**Parking Seeking:**

- Confirmation of match. Example: "Your match for parking at [Location] is confirmed. Head over to the spot!"
- Navigation link to the parking spot. Example: "Navigate to your parking spot using this link: [Map Link]."
- Reminder when hosting period is about to end. Example: "The hosting period for your matched spot will expire soon. Please proceed to the location."

**Parking Swapping:**

- Notifications for arrival and readiness confirmation. Example: "The seeker has arrived at the spot. Please confirm the swap completion."
- Reminder to confirm swap completion. Example: "Don’t forget to confirm the swap once complete!"

**Disputes:**

- Acknowledgment of dispute submission. Example: "Your dispute has been submitted successfully. We are reviewing it."

- Updates on automated or manual review status. Example: "Update: Your dispute is under automated/manual review."
- Final resolution notification. Example: "The dispute has been resolved. Check the details in your dispute log."

**Ratings:**

- Prompt to rate the other party after successful swap or dispute resolution. Example: "Rate your recent swap experience with [User Name] to help improve the community."

***Notification Settings:***

- Notifications feature cannot be disabled for MVP to ensure critical alerts are received.

5. **Historical Logs**

***Features***

- Users can access logs of:
- **Past Communications:** Records of all preset messages exchanged during parking interactions, with timestamps for transparency.
- Example: "Message sent on 24-Dec-2024, 10:30 AM: 'I have arrived earlier than expected.'"
- **Ratings Given and Received:** Summarized records of ratings exchanged after each transaction, including feedback where applicable.
- Example: "Rating for [User Name] on 20-Dec-2024: 5 stars with feedback: 'Great experience, the spot was perfect.'"
- **Disputes Submitted and Their Resolutions:** Detailed logs of disputes, including reasons, status updates, and final outcomes.
- Example: "Dispute ID: #12345, Reason: 'Spot was occupied,' Submitted: 15-Dec-2024, Resolved: 17-Dec-2024, Outcome: Refund of 2 tokens to Seeker."
- **Comparison with Match Details Page:**

***Key Differences***

- **Timeframe:**
- **Match Details Page:** Temporary and context-limited.
- **Historical Logs:** Long-term and account-wide.
- **Purpose:**

- **Match Details Page:** Immediate reference for ongoing or recent transactions.
- **Historical Logs:** Archival reference for trends, disputes, or personal records.
- **Scope:**
- **Match Details Page:** Single interaction focus.
- **Historical Logs:** Aggregated data from all interactions.
- **Match Details Page:** Context-specific and focuses on single, active parking interactions with real-time updates and actions.
- **Historical Logs:** Broader overview of all past activities and their outcomes, serving as a long-term reference.

***Data Retention:***

- Logs will be stored for a period of 12 months. This duration ensures users have access to sufficient historical data for reference, such as resolving disputes, reviewing past interactions, and analyzing ratings trends. It strikes a balance between usability and data management.
- **Why 12 months?**: This period covers the typical yearly cycle for most user activities and behaviors. For instance, users can review parking trends or disputes over seasonal fluctuations.
- **Can it be shorter?**: For apps with high user turnover or frequent transactions, a 6-month retention period could suffice. However, shorter durations may limit the ability to resolve long-term disputes or analyze trends effectively.
- Older records will be archived or deleted as part of periodic housekeeping.
- **Photos and Data-Intensive Information**
- Data-intensive information, such as photos, will be kept for 24 hours.
- If there is no dispute within 24 hours, the data will be automatically deleted from the database.
- If there is a dispute within 24 hours, the data will be retained for an additional 48 hours.
- After the additional 48 hours, the data will be permanently deleted from the database.
- The 24-hour retention period starts from the date and time of the completed transaction or the transaction being canceled.